

Predicting Water Potability: A Comparative Study of Decision Tree vs. Random Forest

Motivation and Description

- Create Decision Tree (DT) and Random Forest (RF) classification models using supervised learning to predict water potability
- Comparing and analysing both the models through evaluation metrics such as F1-Score, Test Accuracy and AUC and investigate which model give optimal results in predicting water potability.

Initial Data Prepossessing and Analysis

- The Water Potability Dataset is from Kaggle, created by Aditya Kadiwal
- The original dataset consists of 3,270 rows, 9 features, 1 target which is binary (0 = Not Potable and 1 = Potable)
- The original dataset had 1,434 missing values therefore I dropped 1,265 rows which contained these missing values
- Table 1 shows the descriptive statistics which clearly shows that all the variables apart from *Solids*, has a Gaussian distribution since their mean and median are extremely close, while for *Solids* the mean is greater than the median value, indicating a right skew.
- Figure 1 shows the class imbalance in the target variable, the cleaned dataset has a 59.7% (Not Potable) / 40.3% (Potable) split.
- Figure 2 shows the feature distributions, the reflections from Table 1 are visually emphasised here. The bell-shaped curve of the histograms indicates that all features have a Gaussian distribution apart from *Solids* which is right-skewed.
- A Z-score normalisation was applied on the features to standardise the varying scale differences, as shown in Table 1.
- Figure 3 shows the Pearson's correlation matrix where all the coefficients are <0.2, this indicates an extremely weak linear relation between all the features. This confirms the features are independent and multicollinearity isn't an issue.

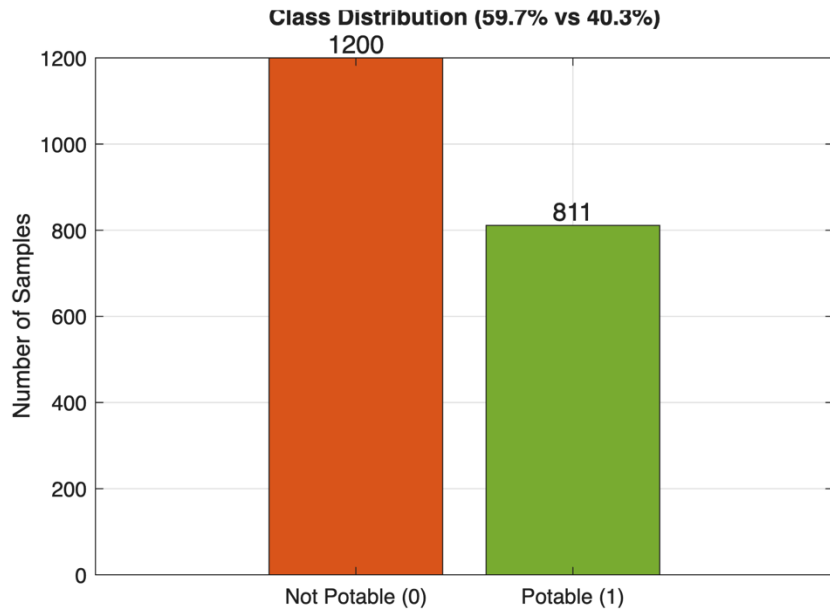


Figure 1: Class distribution of Potability

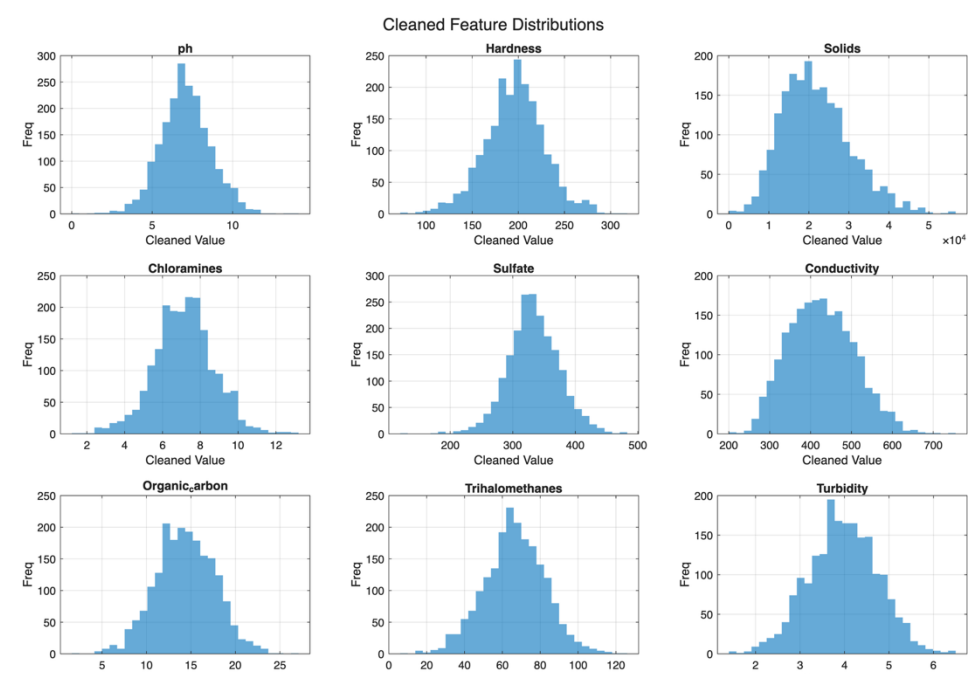


Figure 2: Histograms

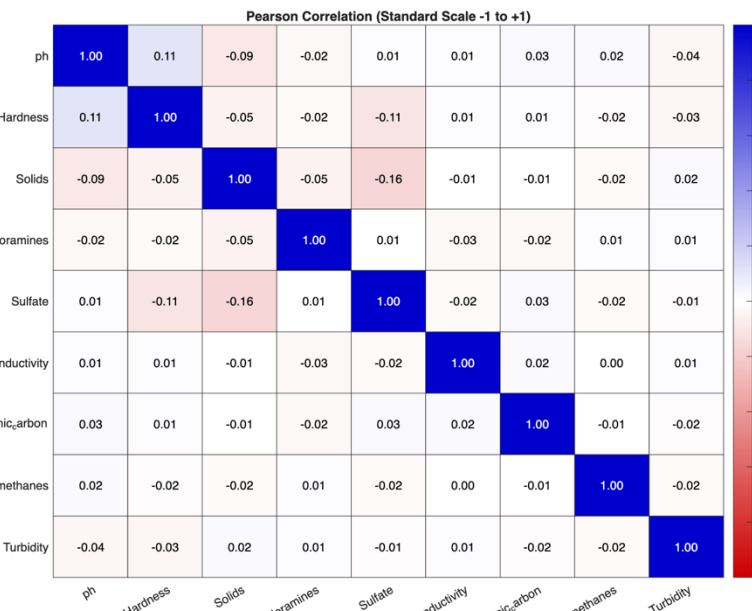


Figure 3: Correlation Heatmap

Features	Mean	Median	StDev
ph (0 to 14)	7.086	7.0273	1.5733
Hardness (mg/L)	195.97	197.19	32.635
Solids (ppm)	21917	20934	8642.2
Chloramines (ppm)	7.1343	7.1439	1.5848
Sulfate (mg/L)	333.22	332.23	41.205
Conductivity (µS/cm)	426.53	423.46	80.713
Organic Carbon (ppm)	14.358	14.322	3.325
Trihalomethanes (µg/L)	66.401	66.542	16.077
Turbidity (NTU)	3.9697	3.9682	0.78035

Table 1: All the features

Hypothesis Statement

- Previous studies have shown that while Decision Trees offer high interpretability for water quality parameters (Saghebian et al., 2013), they are prone to overfitting. Ensemble methods like Random Forest have recently demonstrated superior accuracy (98%) in similar water quality tasks (Shofiana & Caniadi, 2025).
- The Random Forest model is likely to have a much longer test time than the Decision Tree Model.
- The high volatility, skewness of *Solids* will cause instability in the Decision Tree, while the Random Forest model will be resilient to this due to its bootstrapping technique.

Training and Evaluation Methodology

- A stratified split of the dataset was conducted where the dataset was randomly separated into a Training Set (70%) and Testing Set (30%).

Model Optimisation Strategy

- Decision Tree**
 - A 5-fold cross validation on the 70% training set was implemented to estimate generalisation performance; the result is used to make sure of a robust hyperparameter tuning
 - A systematic hyperparameter optimisation grid search method was implemented to prevent overfitting and to optimise the tree complexity, the minimum number of samples required in a leaf node (MinLeafSize) was tuned in the range 1-150.
- Random Forest**
 - An OOB (Out of bag) estimation was used which utilises the leftover bootstrap samples to evaluate the model as it develops, therefore providing a statistically valid and efficient method of model evaluation without reducing the training dataset.
 - A systematic hyperparameter optimisation grid search method was implemented to prevent overfitting; to optimise ensemble size, the number of individual trees (n_estimators) was tuned in the range 50-200.

Evaluation Metrics

- Since there is a significant class imbalance (Figure 1), along with test accuracy, the F1-Score and AUC-ROC were also implemented to evaluate how well the model was able to minimise false negatives and manage the trade-off between precision and recall.

Analysis and Critical Evaluation of Results

- The Random Forest model achieved a better test accuracy (68.82%) shown in Table 2, compared to the Decision Tree model (63.52%). This was because the Random Forest was able to reduce variance by using an ensemble approach. By using 200 independent estimators and applying Bootstrap Aggregation to them, the model separated the signal from the noise associated with the training data, this aligned with Brieman's foundational theory that bagging is essential for stabilising high-volatility estimators (Breiman, 2001).
- The significant difference in AUCs (DT = 0.63 vs RF = 0.72) in Table 2 indicates that the rigid boundaries of the Decision Tree failed to capture the subtle variations, which are captured in the Random Forest due to its monotonic ranking. The ensemble's structural superiority makes it the better candidate for real-world application than the Decision Tree. This finding is also aligned with Shofiana & Caniadi (2025) who confirmed that in assessing water quality, ensemble methods consistently demonstrate superior probability calibration compared to single estimators.
- In terms of the F1 score displayed in Table 2, the Decision Tree (0.54) achieved a higher score than Random Forest (0.47) through its greedy search to maximise recall which resulted in an unacceptably high number of false positives (Figure 4). This emphasises the importance of minimising false positives to avoid public health risks, thus exposing the operationally unsafe nature of the Decision Tree's approach (Chen et al., 2024).
- The Random Forest focuses on precision, using more, using a more conservative classification threshold and minimizing false positives (shown in Figure 5). This risk profile suggests that the Random Forest is preferable for implementation as it fits with the assessment requirements identified by Chen et al. (2024) around the seriousness of Type I errors (e.g., approving contaminated water as safe) as opposed to Type II errors (e.g., discarding safe water).
- The Random Forest's relatively low F1 score of 0.48 (Table 2) indicates that it can be significantly biased by "Majority Voting" when trained on datasets with a high degree of skewness toward one class (59.7% are Not Potable) since the dominant class will effectively mask or suppress any signals from the minority class. This observation is consistent with what Chen et al. (2024) attribute as a common issue with the traditional usage of bagging algorithms to address the effects of majority bias in environmental datasets.
- While Random Forests had a computational burden that was approximately 77 times that of using Decision Tree (5.54 seconds for RF and 0.07 seconds for DT), with regards to batch-processing water quality data, the associated latency with using RF is essentially unimportant. As Saghebian et al. (2013) indicated, the most important aspects of environmental monitoring predictions are accurate assessments of risk and health concerns, rather than speed of real-time processing; thus, for the 5.3% increase in accuracy and an increased risk profile associated with using RF, the trade-off for added complexity was easily justified.
- Empirical evidence of Random Forest's reliability has been provided through the confusion matrix in Figure 5 shows that Class 0 achieved an accuracy of 91.4% with the random forest, while the decision tree showed a dramatic variance at 71.1% for Class 0. The random forest model demonstrates support for Breiman's assertion regarding the ability of OOB guided bagging to reduce noise to generate a generalizable model, without the need for additional validation sets (Breiman, 2001).

References

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Experimental Results & Discussion

Hyperparameter optimisation outcomes

- Decision Tree:** the best leaf size was 20, determined by 5-fold cross validation.
- Random Forest:** the OOB error determined an optimal ensemble size of 200 trees
- Table 2 displays the performance metrics of the test set for both models, which clearly indicates a distinct trade-off between the two models shown by their difference in F1 score.
- Figure 4 shows that the Decision Tree was able to correctly identify a higher number of potable water, this led to the high F1 score shown in Table 2, despite the low overall accuracy.
- Figure 5 shows that the Random Forest model ensemble seemed to have favoured towards Not Potable as a result achieving a high precision, while being less effective in identifying the Potable class which was smaller in proportion.
- Figure 6 clearly indicates that the Random Forest outperforms the Decision Tree since it has a higher true positive rate and a lower false positive rate, compared to the Decision Tree.

Water Potability Evaluation Metrics Results		
Random Forest	Metrics	Decision Tree
5.5416	Avg. Time (s)	0.071964
82.031	Train Acc.	74.077
68.823	Test Acc.	63.516
0.47778	F1 Score	0.53586
0.71973	Avg. AUC	0.63277

Table 2: Model Performance Metrics

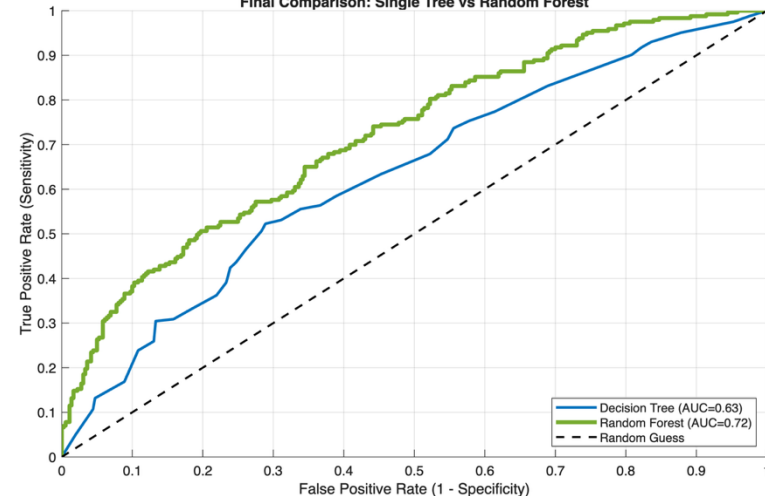


Figure 6: ROC curves of DT and RF

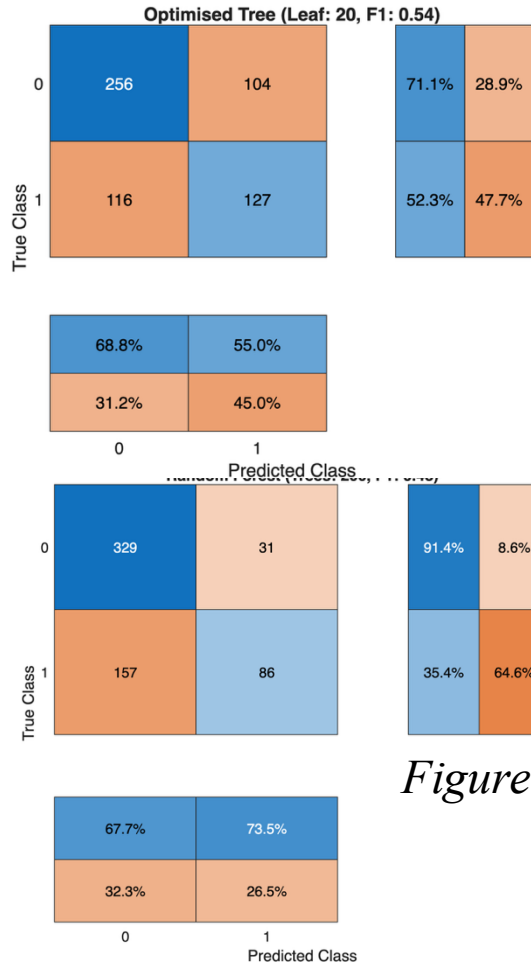


Figure 4: DT Confusion Matrix

Figure 5: RF Confusion Matrix

Model 1: Decision Tree

A supervised learning method that models decisions as a hierarchical tree structure, segmenting the data into smaller groups by splitting the features into binary values and finally allocating a class label to each node at the end.

Model 2: Random Forest

An ensemble method where high magnitude of decision trees is created using a random subset of the data; the results of many trees are merged to increase precision and reduce overfitting.

Decision Tree	Random Forest
Pros: - High interpretability - Less time-consuming when its run	Pros: - Tackles high variance through bootstrapping - Resilient to volatility - Overfitting isn't an issue because it consists of multiple trees
Cons: - Highly prone to overfitting - Can have instability when dealing with highly skewed features like Solids	Cons: - Takes longer time when its run (computationally costly) - Highly sensitive to skewed data

Learning Outcomes

- Ensemble Learning helped stabilised high-volatility estimators, like Solids. the use of ensemble learning provided separation between the true signals and stochastic errors.
- The differences between the Decision Tree's high recall and the Random Forest's higher precision indicated that when evaluating models, the quantitative cost of error (Type I and Type II) must be considered and should not just rely on simply maximisation strategies.
- Bagging techniques had trouble overcoming the bias against the minority class (Potable). The Random Forest model was found to have very high sensitivity to the distribution of the underlying data.

Future Work

- To improve the Random Forest's low F1 score (0.48), the Synthetic Minority Over-Sampling Technique (SMOTE) should be incorporated which effectively mitigates majority bias, showing significant improvement in the sensitivity of classification models to minority classes (Fernandez et al., 2018).
- Instead of relying on the default classification threshold of 0.5, a threshold optimisation method should be implemented where the decision boundary is systematically adjusted to a higher, stricter probability so that the model is optimised to reduce its false positives.
- Instead of dropping the missing values, Multiple Imputation by Chained Equations (MICE) should be implemented to predict missing observations based on their correlation with other features, therefore, generating a more robust model (Buuren and Groothuis-Oudshoorn, 2011)

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